Dohyun (Eric) Kim

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EDUCATION

University of Auckland — **Auckland, New Zealand**
Bachelor of Engineering (Honours) in Computer Systems Engineering — *Expected Graduation Nov 2027*

- **Co-Founder & Academic Team Executive** of *Korean Engineering Body (KEB)* — *Jul 2024 - Oct 2026*

EXPERIENCE

Research Assistant Trainee — **Auckland, New Zealand**
University of Auckland — *Jul 2026 - Oct 2026*

- Trainee researcher analysing the **robot soccer system** at UoA's CARES Lab, covering camera calibration, **vision processing and decision strategy design**, to support its planned migration to a modern framework.
- Investigated **TurtleBot2** navigation algorithms and implemented LiDAR integration and control using **ROS and Python**, building hands-on experience in autonomous systems.

Robotics Instructor — **Auckland, New Zealand**
Creative Imaginary Lab (ciLab) — *Apr 2026 - Oct 2026*

- Instructed **50+ students** in robot **hardware assembly and software programming** to complete missions.
- Supported student development and coached teams in preparation for **nationwide robotics competitions**.

PROJECTS

Sentiment PULSE — **Live Demo**
Real-time Sentiment Dashboard - Personal Project — *Jun 2026*

- Built a serverless **streaming sentiment pipeline** on **AWS (Kinesis, Lambda, Comprehend, DynamoDB)** using SAM, with idempotent writes, partial batch retries and a DLQ to prevent data loss.
- Replaced full table scans with a **bounded DynamoDB GSI query** to keep read latency low, and batched **Comprehend calls per Lambda invocation** to cut API overhead.
- Built a live React dashboard on AWS Amplify with a client-side simulation fallback and **GitHub Actions CI (21 frontend, 8 backend tests)**, applying **AWS certification fundamentals** hands-on.

SLOTLY — **PostgreSQL**
Lab Equipment Booking System - Personal Project — *Oct 2026*

- Built a lab equipment booking REST API with **Express.js and PostgreSQL**, using an exclusion constraint (btree_gist, tstzrange) to make double booking impossible at the database level.
- Wrote **24 integration tests** against a real Postgres, using a 20 request concurrency test to expose a deadlock and row level locking to fix it, cutting failures from 5 in 10 runs to 0 in 20.
- Defined the API's **AWS infrastructure (VPC, RDS, EC2) in Terraform**, applying my Terraform Associate certification, and packaged it as a **non root Docker** image with GitHub Actions CI.

AWARDS

Top 8 Finalist in AWS×BNZ AI Hackathon 2026 — **Live Demo**
"Spottern!" - Statement-level Fraud Detection Platform — *Jul 2026*

- Built a serverless fraud pipeline with **AWS (S3, Lambda, Textract, Bedrock, DynamoDB)**, using Claude Opus 4.8 on Bedrock **to extract and flag anomalies from bank statements** with cited transaction evidence.
- **Proposed the approach to BNZ** by drawing on prior personal project experience with serverless AWS pipelines, placing **top 8 out of 20+** teams.

CERTIFICATIONS

AWS Certified Solutions Architect - Associate · Amazon Web Services — *Aug 2026*
HashiCorp Certified: Terraform Associate · HashiCorp — *Sep 2026*
AWS Certified AI Practitioner · Amazon Web Services — *May 2026*
AWS Certified Cloud Practitioner · Amazon Web Services — *Apr 2026*

SKILLS

Languages & Tools: Python, C/C++, Java, JavaScript, TypeScript, SQL, R, HTML/CSS, Git, Android Studio, Figma
Frameworks: React.js, Tailwind CSS, Node.js, Express.js, PostgreSQL, JUnit, ROS, REST APIs, Spring Boot
Cloud & DevOps: AWS (Lambda, DynamoDB, S3, Bedrock, Kinesis), Docker, Terraform, GitHub Actions